AI Adoption Initiative Design Canvas

AI-Assisted Review for Senior Engineering Design Projects

Module 6 · Task 1 | Rey Granillo III | AI Campus Champions, Summer 2026

SECTION 1 — Problem Statement

Engineering senior design mentors must review lengthy technical reports for the college's **3 capstone teams (~20 students)** across two semesters on tight timelines. Feedback depth varies team to team, and project gaps slip through to Design Day. This initiative uses AI to give mentors consistent, targeted review support at each milestone.

SECTION 2 — Target Audience

Role(s)	Faculty and staff researchers serving as senior engineering design mentors
AI fluency (1–5)	2–3
What they need	A streamlined, semi-automated review process that directs them to the areas worth focusing on within long technical reports, without adding an approval or tooling burden.
Key concern	Credibility of AI-generated reviews. The design meets this by using chain-of-thought (CoT) prompting, which elicits explicit intermediate reasoning steps the mentor can inspect [Wei et al., 2022] rather than trusting an opaque verdict — a direct check against hallucinated or source-unfaithful output in generated text [Ji et al., 2023].

SECTION 3 — AI Tools and Approach

Tool	Modality	Rationale (evidence-grounded)	Data Sensitivity	Approved?
Claude (via UA GenAI), using CoT	Text generation / reasoning	CoT surfaces explicit reasoning steps a mentor can verify [Wei et al., 2022], mitigating hallucinated output [Ji et al., 2023]; used within UA GenAI to keep the workflow FERPA-compliant. Fits AI-assisted (not AI-replaced) review inside the ‘jagged frontier’ where AI reliably lifts knowledge work [Dell'Acqua et al., 2023].	Low — projects are built to be scaled and typically open-sourced	Yes
GitHub Copilot	Automated code review	Native to GitHub, where project code already resides; likely one- or zero-shot interactions for repository-level review.	Low	Pending — Microsoft Copilot is on UA's Responsible AI list, but GitHub Copilot is not explicitly listed; to be confirmed

SECTION 4 — Ethical Safeguards

Risk	Safeguard Mechanism	Responsible Party
Disclosure of protected data — some projects involve regulated, confidential, or identifiable data	Policy education on what is safe to share; identifiable/protected data is never entered into any AI tool; sensitive work restricted to approved UA GenAI	Researchers and students providing the data
Student-record privacy in persona tuning (FERPA)	Only de-identified, aggregated example excerpts are used to build or refine the reviewer persona; identifiable grades, names, or records are never entered into any tool	Initiative lead + department data/FERPA liaison
Over-reliance / unverified output — treating a fluent model as authoritative	Mandatory human-in-the-loop review of every AI-surfaced finding via CoT before it reaches a student; the model surfaces, the mentor decides [Bender et al., 2021; NIST AI RMF, 2023]	Mentors
Intentional misuse	AI-use and disclosure policies; adherence to documented best practices	Model users (mentors and students)

SECTION 5 — Equity Considerations

Population	Barrier	Adjustment (population-specific)
Students with limited or unreliable technology / internet access	May be unable to access AI tools to review flagged feedback promptly, widening the gap between teams	Mentors deliver AI-surfaced talking points directly in scheduled in-person check-ins, so tool access is never a prerequisite; use only the free, institutionally hosted UA GenAI rather than paid tiers
Multilingual and international students	Technical-writing feedback and AI phrasing can disadvantage non-native English writers, and AI-text detectors disproportionately misflag them	Mentors frame AI output as engineering-content and structure guidance, not language policing; explicitly separate 'is the engineering complete?' from 'is the prose native-fluent?' [fairness concern per Barocas, Hardt & Narayanan]
Mentors with lower AI fluency (2–3)	Uneven ability to steer the tool with CoT produces uneven feedback quality across teams	Provide a shared prompt template and a worked example from AI Champions training, so review quality does not depend on an individual mentor's skill — knowledge stewarding within a community of practice [Wenger, McDermott & Snyder, 2002]

SECTION 6 — Success Metrics

Metric	How Measured	Timeline	Minimum Threshold
Student progress	Faculty-advisor check-ins across scheduled milestone presentations; graded critiques against the cognitive levels each milestone should assess [Krathwohl, 2002]	Pre-set group presentation dates	All required points addressed at each presentation
End-of-project / final review	Final Design Day presentation + report, assessing whether AI-informed recommendations were correct / well implemented	End of two semesters	Passing grade — B or above

SECTION 7 — Rollout: First Three Steps (90-Day Pilot)

Step	What Happens	Who / Resource	By When
1 · Days 1–30	Build a prompt-based expert-reviewer persona (instructions + a few de-identified example excerpts) in UA GenAI, with equity and bias-minimizing safeguards baked into the prompt. Executable now, no new permissions.	Me; UA GenAI	Day 30
2 · Days 31–60	Assemble a small validation set of de-identified past-report excerpts (no identifiable grades or PII); run the persona on 2–3 samples and compare its flags against mentor judgment.	Me; de-identified sample reports	Day 60
3 · Days 61–90	Pilot the persona live at one milestone checkpoint with a small group of volunteer mentors; collect structured feedback and refine the prompt.	Me + volunteer mentors	Day 90

Scope note: iterative refinement across a full project cycle — and any move beyond prompt-based use toward a tuned model — is future work ***outside*** *this 90-day pilot, and would require a separate FERPA data-use review before any student-record data is aggregated.*

30-Second Elevator Pitch. *Senior Design mentors face lengthy technical reports with limited time. Our initiative uses Claude's chain-of-thought reasoning to flag incomplete or problematic areas and generate targeted talking points—keeping faculty in control. The result: faster, more consistent, credible reviews that help engineering students strengthen their capstone projects before Design Day.*

References

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